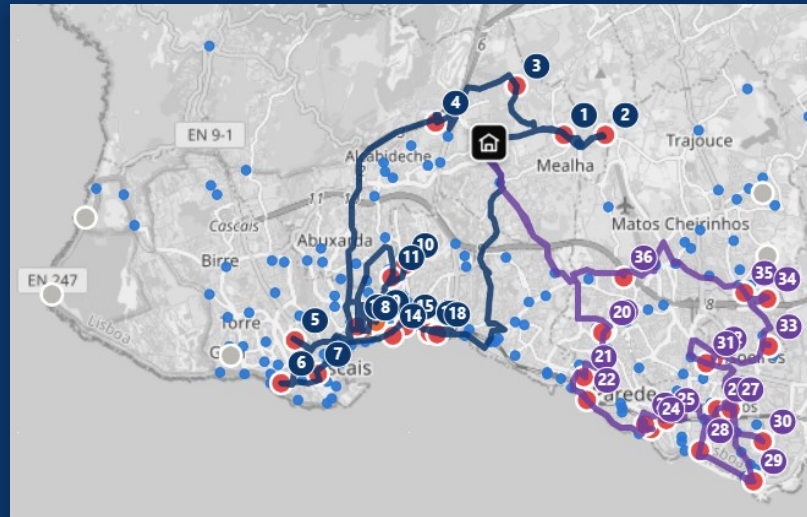


Cascais Sweep

Find abandoned scooters.
Plan the van.
Clear the streets.



Collection runs planned on real streets

The problem

Bird (private operator) must keep its scooters at stations. If one stays > 30 m outside for > 2 h, Cascais Próxima (the city) removes it and Bird pays a €4.5 fine. 3 weeks of Bird data:

44

new abandoned
vehicles every day

13.6 h

median time on the street
after the 2 h limit

~10

collected by the city per day
in our data period (one run)

16,888 trips · 1,388 vehicles · 157 stations · 920 abandonment cases · 24,791 vehicle-hours abandoned

Who is affected



Pedestrians

30 cases blocked a bus stop (within 20 m) for 657 hours in total — wheelchairs, prams, visually impaired people.



Cascais Próxima team

One van, 18 places, no tool: which vehicles matter most, and in which order to drive?



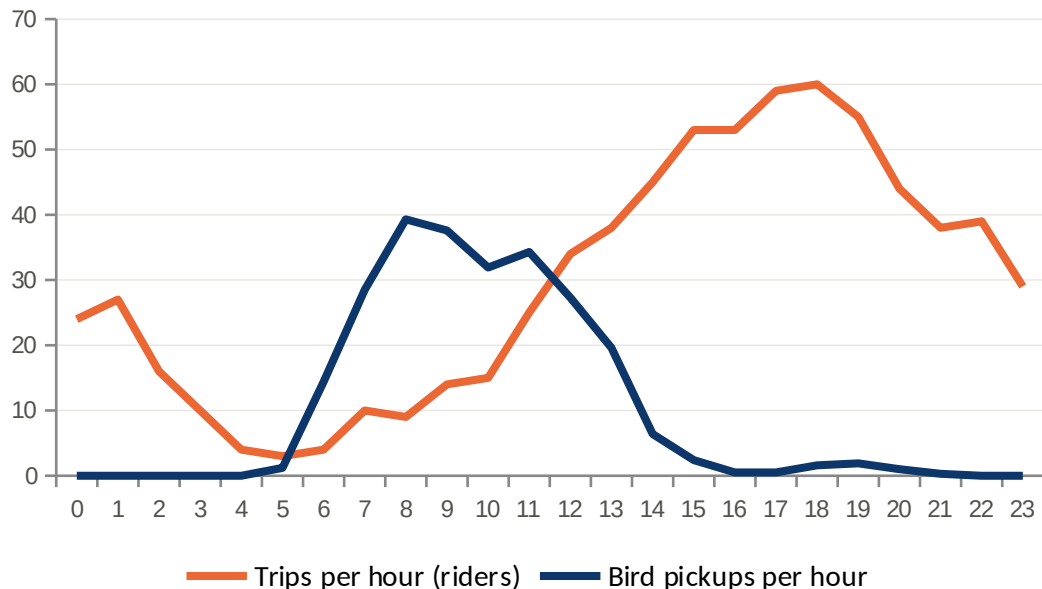
The city vs the operator

Two different organisations: Bird earns from rides, the city keeps the streets clear. Every scooter the city removes costs Bird €4.5.

Rules designed with João Silva, Cascais Próxima: depot, van capacity, how the team works, the fine.

Why they stay so long

Average per day, by hour



94%

of Bird pickups happen 06:00–14:00. Riders peak in the evening, so vehicles left after lunch wait until morning.

109

times a Bird technician serviced an abandoned vehicle on the spot and left it there: 6,652 more hours.

6–14%

of cases would disappear even with the 8 best new stations. Abandonment is everywhere: collect, don't build.

Detect → Prioritise → Route



1. Detect

History: 3 weeks of Bird events.
Live: Bird's public GBFS feed, every minute. Vehicles matched between snapshots by position.



2. Prioritise

Time abandoned, distance from a station, blocking a bus stop, and a model: will a rider take it anyway? Weights are sliders.



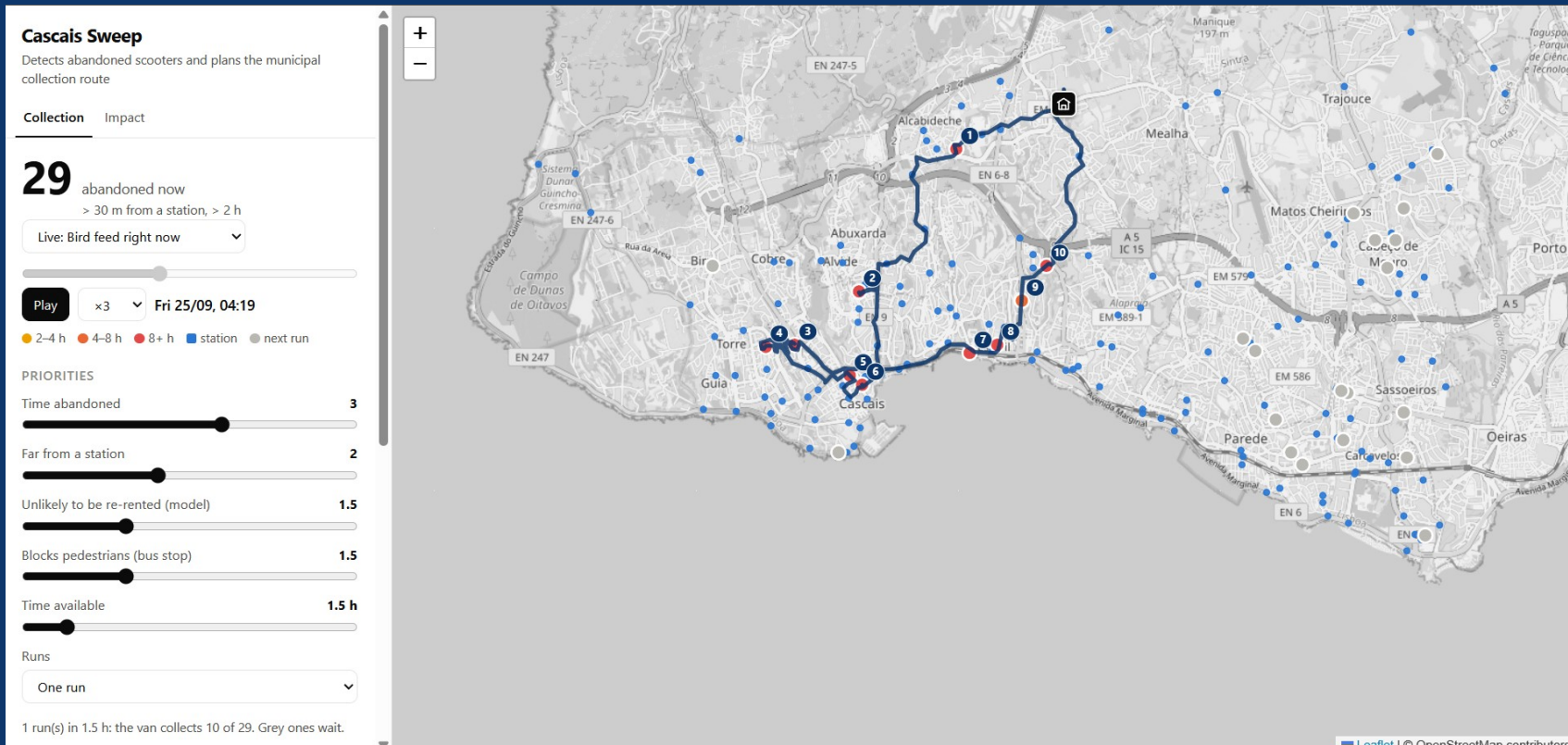
3. Route

Pick the vehicles worth most per minute of detour, within 18 places and the time available. Order and path on real streets (OSRM).

Open source only: Python, Leaflet, OpenStreetMap, OSRM. No API keys.

Live demo

github.com/Artem-Tkachev/cascais-sweep



Impact

Simulation on 3 weeks of real data: one van, weekdays 09:00–18:00, 18 vehicles per run

–59%

vehicle-hours abandoned
24,791 → 10,246

30.7

vehicles a day
instead of ~10 in our data

–30%

km driven vs first-come
order
94 vs 134 km a day

+€2,000

finer a month
€45 → €138 a day

In one 2-hour run: 12.7 vehicles instead of ~10. Vehicles at bus stops collected first.

What's next



Any operator, any city

Built only on GBFS, the open standard of shared mobility.



Pilot with Cascais Próxima

Live feed on a small server; store events instead of snapshots.



Smarter multi-run plans

A VRP solver (VROOM on OSRM) to balance vehicles between runs.



Where are buses missing?

Scooter trips from bus stops show where and when people need new bus links. Cascais Próxima already shared its stops and routes data.